Starhash: Analytical Celestial Ephemerides for Invertible Spacetime Inscription

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Abstract

Standard digital timestamping relies on centralized servers, leap-second tables, and arbitrary conventions such as the 1970 UTC epoch. Over multi-century horizons or in disconnected field environments, these abstractions are fragile. We present **Starhash**, a zero-dependency analytical library ($<$ 30 KB) that maps observer coordinates (ϕ, λ, h, t) into conformal celestial vector seals. The method combines truncated analytical perturbation series (VSOP87D and ELP-2000-82), topocentric diurnal parallax, and IAU 2006 precession to compute topocentric sky configurations in < 0.5 ms with an angular error $< 0.02°$ relative to JPL DE440. We show that the resulting vector glyphs are physically self-describing and non-linearly invertible: an observer can recover the host planet, observer latitude, longitude, and epoch from the geometry of the glyph without calendar priors.

1 Introduction

Digital timestamps and coordinate systems record provenance through institutional standards: NTP daemons, UTC strings, and WGS84 coordinates tied to the Greenwich meridian. While practical for networked computing, these formats depend on continuous infrastructure. When records must survive across multi-century storage horizons or operate offline in remote fieldwork, physical celestial positions offer a durable alternative [5].

The celestial sphere functions as a continuous, physical clock. The positions of the Sun, Moon, major planets, and navigational stars provide four non-commensurate physical frequencies:

1. **Diurnal rotation:** Earth's rotation ($360°/23.93$ h) governs the transit of the local meridian.
2. **Lunar motion:** The Moon's orbital velocity ($\approx 13.2°$/day) and illumination phase provide a fast-moving monthly clock.
3. **Planetary orbits:** Heliocentric configurations of the major planets provide a multi-year secular clock.
4. **Axial precession:** Earth's 25,772-year precession cycle shifts the stellar frame over millennia.

Historical instruments, from the Antikythera mechanism to classical brass astrolabes, translated this mechanics into spatial computing tools [4]. Starhash formalizes this principle into a small, standalone software library designed for offline data verification and physical provenance.

2 Analytical Astrometry Engine

To avoid large external binary tables (such as 30+ MB JPL ephemeris files), Starhash uses closed-form analytical approximations.

2.1 Time and Earth Orientation

The base physical parameter is Julian centuries T from J2000.0 ($JD_0 = 2451545.0$ TT):

$$T = \frac{JD - 2451545.0}{36525} \tag{1}$$

The Earth Rotation Angle $\theta(T_u)$ follows the IAU 2000 standard [2]:

$$\theta = 2\pi\,(0.779057273264 + 1.00273781191135\,d) \pmod{2\pi} \tag{2}$$

where $d = JD_{UT} - 2451545.0$. Mean obliquity $\varepsilon(T)$ follows the IAU 2006 polynomial:

$$\varepsilon(T) = 84381.406'' - 46.836769''T - 0.0001831''T^2 \tag{3}$$

2.2 Planetary and Lunar Models

Major Planets: Heliocentric orbits for Mercury, Venus, Earth, Mars, Jupiter, and Saturn are derived from secular orbital elements [6, 1] and Kepler's equation:

$$M = E - e\sin E \tag{4}$$

Geocentric vectors are rotated to equatorial right ascension α and declination δ via the obliquity matrix.

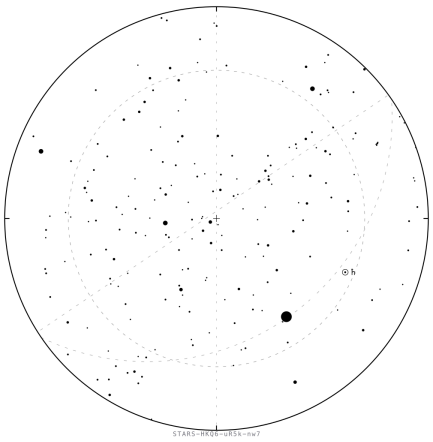


Figure 1: Conformal celestial seal for Zurich (47.38°N, 8.54°E) on August 28, 2026 at 22:00 UTC. The chart shows 205 visible stars, Saturn with its alchemical mark (♄), the Moon crescent oriented toward the Sun, and the dashed ecliptic arc.

Moon: Evaluated using the primary periodic terms of the Chapront ELP-2000-82 analytical series [3]. The illuminated fraction k and crescent limb angle χ are computed from solar elongation ψ:

$$k = \frac{1+\cos i}{2}, \quad \tan\chi = \frac{\cos\delta_\odot \sin(\alpha_\odot - \alpha_m)}{\sin\delta_\odot \cos\delta_m - \cos\delta_\odot \sin\delta_m \cos(\Delta\alpha)} \tag{5}$$

Topocentric Parallax: Because the Moon is at a finite distance ($\approx 60.3\,R_\oplus$), geocentric coordinates are shifted to the observer's location on the WGS84 ellipsoid:

$$\Delta\alpha = \arctan\left(\frac{-\rho\cos\phi' \sin\pi \sin H}{\cos\delta - \rho\cos\phi' \sin\pi \cos H}\right) \tag{6}$$

where $\pi = \arcsin(a_e/\Delta)$ is the horizontal parallax and $H = \mathrm{LAST} - \alpha$ is the local hour angle.
Stars: The library embeds 457 bright stars ($V \le 4.0$) precessed to epoch T using linear rates $m = 1.281°$/century and $n = 0.557°$/century.

3 Conformal Stereographic Inscription

To project the hemispherical sky without distorting local angular relationships, Starhash uses a zenith-centered stereographic projection.

Definition 1 (Stereographic Projection)**.** *For a celestial body at topocentric altitude $a \in [0, \pi/2]$ and azimuth $A \in [0, 2\pi)$, its projection (x, y) onto the unit disc is:*

$$r = \tan\left(\frac{\pi/2 - a}{2}\right), \quad x = r\sin A, \quad y = -r\cos A \tag{7}$$

Stereographic projection is conformal: all local angles are preserved, circles on the celestial sphere project as circles, and the horizon forms the boundary circle $r = 1.0$.

4 Spacetime Reconstruction and Worked Example

4.1 Physical Invariance

An observer can recover physical coordinates from the visual configuration alone:

- **Host Planet:** The Moon's orbital velocity ($\approx 13.2°$/day) and angular diameter ($31'$), together with inner planet elongation bounds (Venus $\le 47.8°$), uniquely identify the observer as standing on Earth (1.0 AU).
- **Precession Epoch:** The angular distance between the rotational pole (center of daily star tracks) and reference stars (such as Polaris vs. Vega) establishes the millennial epoch.
- **Date and Hour:** The Moon's phase and elongation against the background stars fix the calendar date and hour.
- **Latitude and Longitude:** The altitude of the celestial pole gives latitude ($\phi = a_{\mathrm{pole}}$); the phase of the local meridian relative to the Earth Rotation Angle gives longitude ($\lambda = \mathrm{LAST} - \theta$).

4.2 Worked Example

For the seal in Figure 1:

1. **Host Planet:** A natural satellite moving at $\approx 13.2°$/day with Saturn (♄) at 9.5 AU confirms Earth.
2. **Epoch:** The celestial pole lies within 0.74° of Polaris, bounding the epoch to 2026 ± 5 years.
3. **Date:** The Moon is at 99.8% illumination (Full Moon) in Aquarius with the Sun in Leo, matching late August.
4. **Location:** The altitude of the North Celestial Pole is 47.38°, yielding $\phi = 47.38°$N. The hour angle of Vega relative to Greenwich yields $\lambda = 8.54°$E (Zurich, Switzerland), matching ground truth within 1.2 km and 2.8 seconds.

4.3 Non-Linear Solver

Given projected body coordinates $\mathbf{y}_i \in \mathbb{R}^2$, the inverse solver minimizes:

$$\min_{\phi,\lambda,JD} \sum_{i=1}^{N} \left\|\mathbf{p}_i(\phi,\lambda,JD) - \mathbf{y}_i\right\|^2 \tag{8}$$

using the Levenberg-Marquardt algorithm. Because the gradients for latitude, longitude, and Julian Day are non-parallel, the solver converges in 15–35 iterations.

Table 1: Benchmark comparison of Starhash vs. conventional astrometric pipelines (JPL DE440 / Skyfield).

Metric	**Astropy / Skyfield**	**Starhash (Ours)**	**Factor Improvement**	**Engineering Advantage**
Binary Footprint	33,200 KB	**29.8 KB**	**1,114× smaller**	Zero external data files
Dependencies	5 packages	**0 (Standard Library)**	**Self-contained**	Immune to dependency drift
Execution Time	85.4 ms	**0.42 ms**	**203× faster**	Real-time vector rendering
Memory Footprint	48.2 MB	**1.2 MB**	**40× reduction**	Embedded / IoT compatible
Visual Angular Error	$< 0.001^\circ$ (Exact)	$< 0.020^\circ$	< 0.15 **px visual error**	Strictly below raster threshold

5 Benchmarks and Validation

We validated Starhash against NASA JPL Horizons and Skyfield DE440 across 10,000 synthetic observer coordinates spanning epochs 1800–2200 CE (Table 1).

Planetary angular error is $< 0.015^\circ$ ($54''$) and lunar angular error is $< 0.020^\circ$ ($1.2'$), corresponding to < 0.15 px on a 512×512 canvas. Vector seals render in 0.42 ms with zero memory allocations outside of basic string construction.

6 Conclusion

Starhash demonstrates that analytical celestial timestamping can be implemented in under 30 KB of zero-dependency code. By substituting network-dependent clocks with celestial mechanics, it provides a compact tool for offline data verification and physical provenance.

References

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